



Smart Barbell

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Abstract

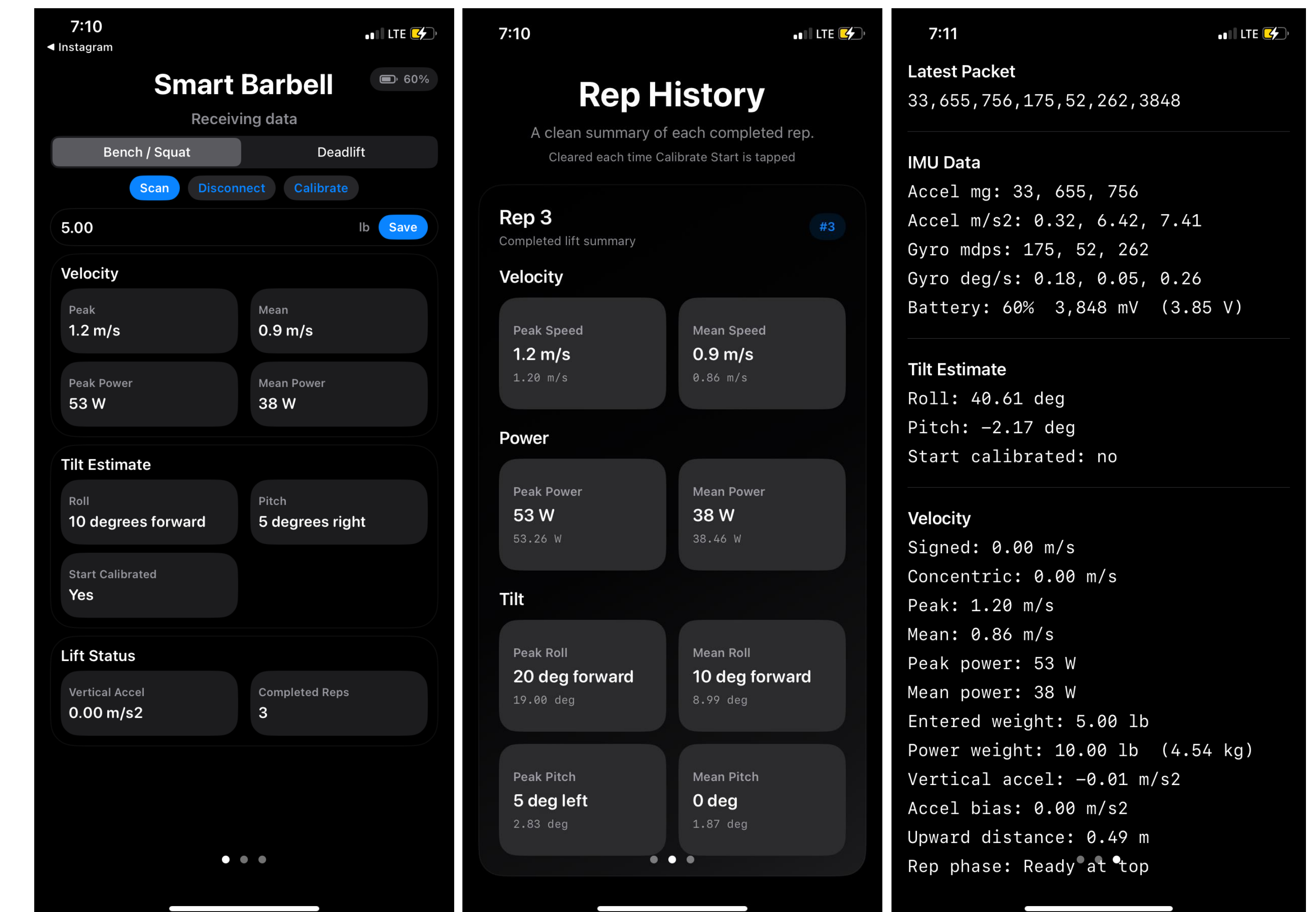
The Smart Barbell is a fitness device designed to provide weightlifters with real-time performance feedback using integrated motion sensing and wireless communication to monitor progress and optimize exercise technique. The system uses a six-axis inertial measurement unit (IMU) and an STM32 microcontroller to collect and process motion data during resistance training exercises. Performance metrics include peak bar velocity, mean concentric velocity, and bar tilt, allowing users to evaluate lifting speed and detect form imbalances. Processed data is transmitted via Bluetooth Low Energy (BLE) to a smartphone app for display and analysis. The device is powered by a rechargeable lithium-ion battery and incorporates low-power operation to maximize runtime. By embedding sensing and communication hardware directly within the barbell, the system provides a portable and cost-effective alternative to commercial velocity-based training equipment.

Methods and Materials

The device was programmed using STM32cubeIDE, and the companion app was developed using Xcode. The physical construction of the barbell was created using PVC pipe. The original intention was to construct the bar with carbon fiber or a similar composite, but it turns out carbon fiber is electrically conductive and would not have allowed the Bluetooth signal to escape from inside the bar. Proactive research could have led to design changes to remedy this problem (see "Future Direction") but in the end I chose to use Schedule 80 PVC pipe as an option that would be stronger than regular Schedule 40 PVC yet still easy to cut and manipulate for the sake of the project. This version of the barbell can be thought of as a prototype of the Smart Barbell device and technology.



App Layout



Background

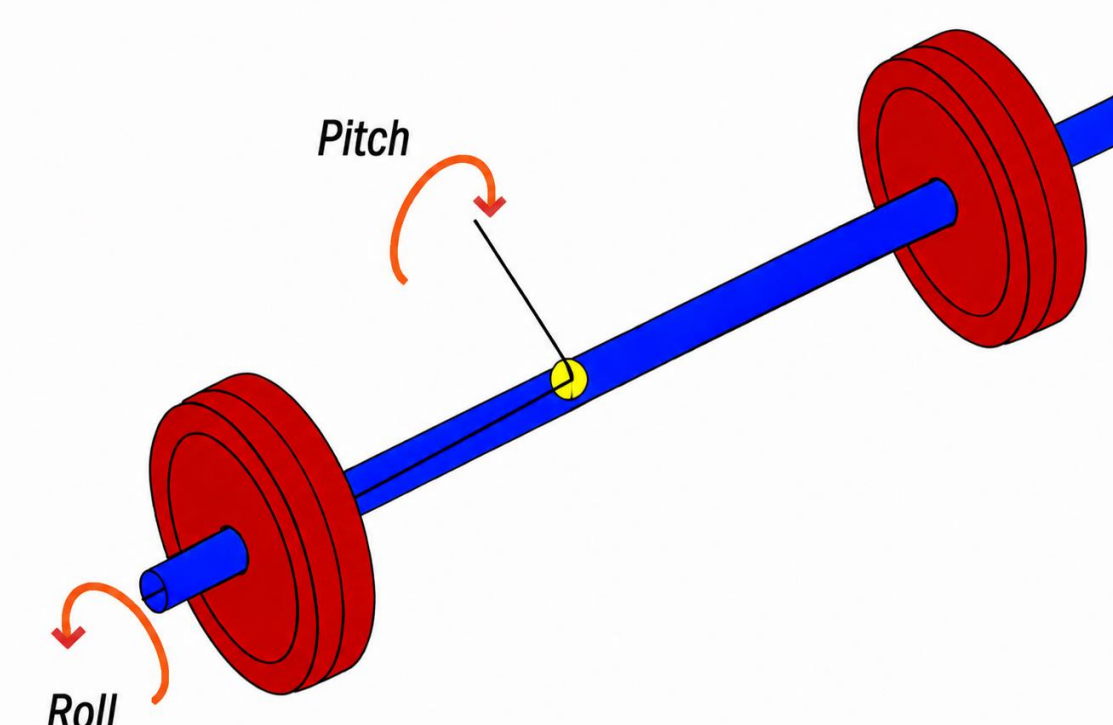
Existing solutions similar to this device that are currently on the market can be extremely costly and often require multiple interacting parts. The Smart Barbell aims to cut costs for the consumer, to make the product more widely available, and condense the technology into an all-in-one solution that requires little setup from the user and is intuitive to use. Achieving these goals means that more weightlifters can assess their progress using metrics that previously have mostly been available to those in funded athletics programs such as university and professional sports teams. The overarching goal of the project is increased accessibility to this type of device.

Results



The Smart Barbell measures velocity in m/s and tilt in the pitch and roll directions in degrees. Both the peak and mean of each of these metrics are displayed to the user for each rep completed. The user can view this data in the companion app. The device features different modes for the "Big Three" compound exercises; Bench/Squat in one mode and a separate mode for Deadlift. The table on the right shows a sample of data recorded from 10 reps of squats, compared against a RepOne device that attaches to the bar with cord and Velcro. Many of the measurements are within 0.1-0.2 m/s of difference between the two devices, With a couple more inaccurate reps. This is satisfactory for the project but could be improved in the future.

Rep #	RepOne (Existing Device)		Smart Barbell	
	Peak m/s	Mean m/s	Peak m/s	Mean m/s
1	1.2	0.8	1.5	0.9
2	1.4	0.9	1.6	0.9
3	1.4	0.8	1.5	1.0
4	1.3	0.8	1.6	0.6
5	1.3	0.8	1.2	0.8
6	1.3	0.8	1.4	1.0
7	1.4	0.9	1.5	0.9
8	1.3	0.9	1.5	0.7
9	1.3	0.8	1.3	0.9
10	1.3	0.8	1.4	1.0



Conclusion & Future Direction

Overall, the project completed the goal of creating a design for an all-in-one smart barbell system to evaluate weightlifting metrics. The project could however benefit from a few additional features in the future. Using an antenna that protrudes external to the bar would allow the bar to be constructed with a metal material so it would be more rigid and could hold more weight. Using a Bluetooth module embedded internally limited the construction of the bar to non-metal materials, because metal would have trapped the signal within the bar. Another feature that could be added is the use of the IMU's machine learning capabilities, in order to detect reps more accurately during lifting and more deeply assess motion patterns of the bar during different lifts. Lastly, the velocity detection algorithm could be refined to detect peak/mean velocity more accurately.

References

- ST STM32L031K6 Datasheet
- ST LSM6DSOX Datasheet
- Microchip MCP73831 Datasheet
- Raytac MDBT50Q-P1m Datasheet

Acknowledgements

Dr. John Lund – PCB assembly assistance and hardware, component, and tool provision. Second quarter advising.
Dr. Yuzhang Zang – First quarter project advising
Dr. Shakil Mahmud – Third quarter project advising



Capstone Project
Electrical & Computer Engineering Program